

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation for safety procedures.

Required Tools

- ◆ Straight-head screwdriver ◆ Torque wrench
- ◆ Cross-head screwdriver ◆ Metric hex-head wrenches

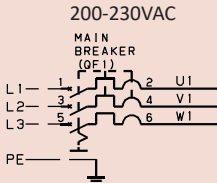
Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label
- ◆ Customer-supplied properly rated input power cable; conductor size of AWG10; terminal size is M5
- ◆ Multimeter ◆ Rigid platform to hold robot
- ◆ 6-M16 x 65mm bolts ◆ Loctite #243
- ◆ 6-M16 high strength washers

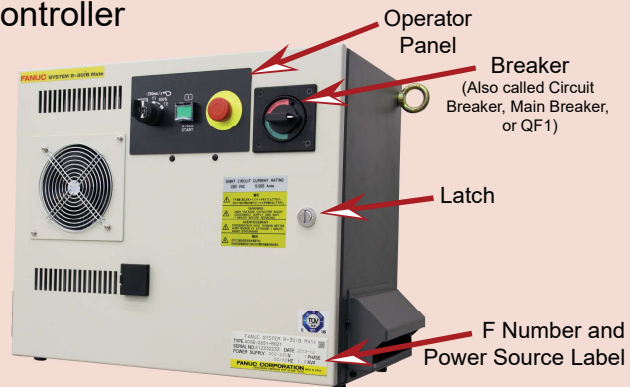
Optional Items

- ◆ Robot locator pins

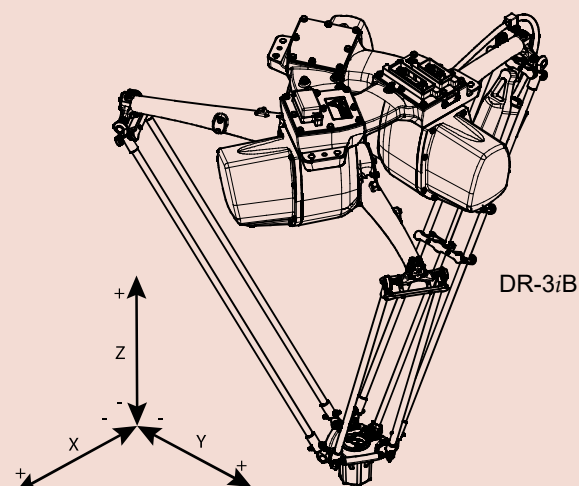
Three Phase



Controller



Robot (mechanical unit)



A. Unpack/Install your Robot



Store these items in a safe, known location.

1. Squeeze and pull the clips off of the skid and remove the cardboard cover. Unwrap the controller and robot. Open all of the boxes included in your shipment.

WARNING

DO NOT disconnect the cable from the remote battery box. Disconnecting the cable can cause the robot to lose mastering data.



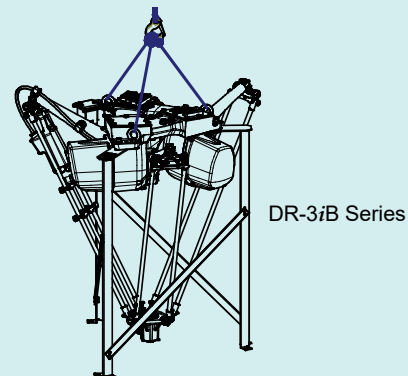
DO NOT disconnect the remote battery cable connection.

2. The robot is mounted to transport equipment on the skid. Remove the bolts holding the transport equipment to the skid.

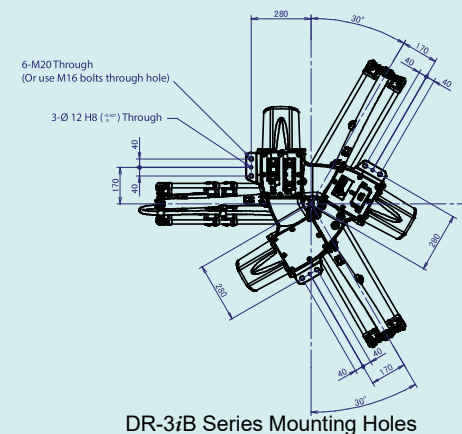
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

3. Use a crane rated for at least 300kg and three slings rated for at least 250kg/sling to move the robot and transport equipment to its new location..



4. Fasten the robot to the fixture with (6) M20 bolts (tightening torque 98NM (10kgfm)). More information is available in the controller specific *Mechanical Unit Operator's Manual*.



5. Uninstall the remote battery box from the base of the robot.

WARNING

DO NOT disconnect the cable from the remote battery box. Disconnecting the cable can cause the robot to lose mastering data.

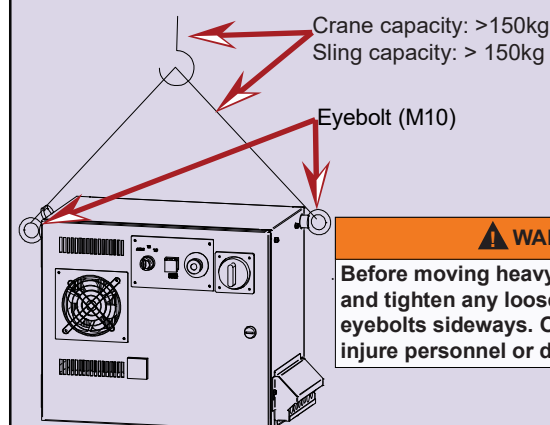
6. Mount the remote battery box in a secure location

Robot installation is complete.

You are now ready to go to **B. Unpack/Install your Controller**

B. Unpack/Install your Controller

1. Use properly rated equipment to remove the controller from its box.



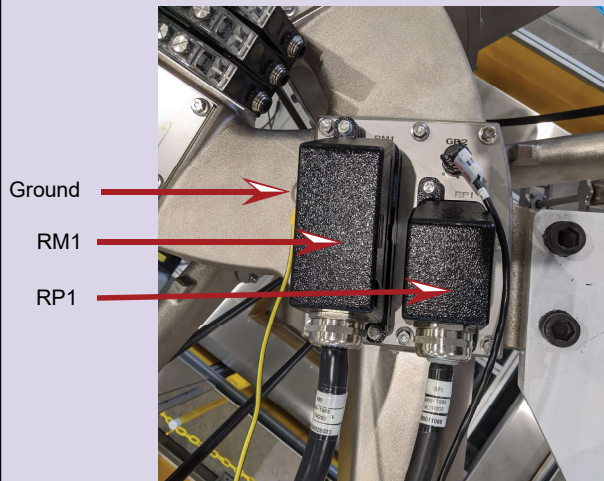
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

2. Place the controller in a safe location where the grounding wire, RM1, and RP1 cables can reach the robot, and the controller is safely reachable from outside of the robot work area.



3. Attach the Robot Connection Cables (RCC) cables (RM1 and RP1) and ground to the robot.



4. Connect the teach pendant to its cable as shown below.

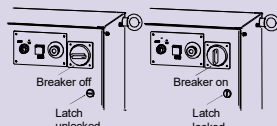


7. Find the power source needed for your controller on the door near the F Number. See diagram on the left side of this Guide for location.

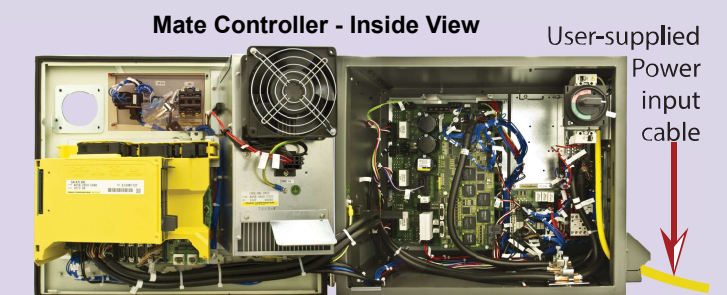
WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

8. Open the controller door by using a straight-head screwdriver to turn the latch counter-clockwise, then turn the Breaker counter-clockwise past its off position.



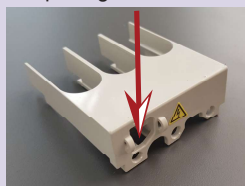
9. Route the power input cable through the side of the controller as shown in the diagram below.



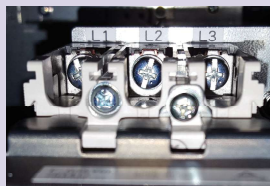
Go to Page 2

10. To connect power, remove the plastic cover from above the breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.

Plastic Breaker cover opening location

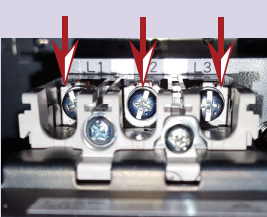


Breaker with cover removed



11. Verify that the input power cable is not energized. Connect power to the breaker with the customer-supplied power input cable. Connect L1, L2, and L3. Connect the ground to the plate to the right of the breaker.

L1 L2 L3

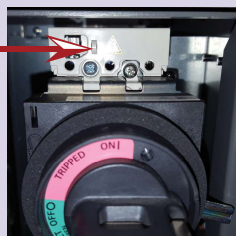


Ground



12. Re-insert the breaker cover and click it into place. Place the breaker access panel back over top of the breaker and secure with four screws.

Breaker with cover reinserted



13. With the breaker still off, energize the input power cable and use a multimeter to verify the correct voltage is present via the holes in the breaker cover.

14. Before you close the controller door, cut loose the set of keys from inside the door and store them in a known location outside of the controller for use later in this guide.

Key location inside controller



15. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

Controller installation is complete.
You are now ready to go to **C. Turn on your Controller.**

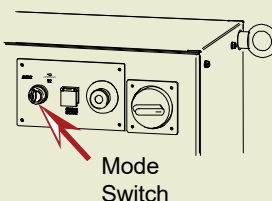
C. Turn on your Controller

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

1. Turn on the controller by rotating the Breaker clockwise to ON.

2. Locate the Mode switch on the Controller Operator Panel. Use the keys retrieved in Step B14 to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.



3. Pick up the teach pendant and look at the screen. Verify that the controller is in T1 Mode as seen in the figure below. If it is not, repeat the previous step and check again.

4. Press the COORD key on the teach pendant until JOINT is displayed



T1 Mode

JOINT



You have turned on your controller.
You are now ready to go to **D. Jog your Robot.**

D. Jog your Robot

1. Turn the teach pendant ON/OFF switch to the ON position.

2. Press **MENU**, move the cursor to **ALARM**, and press **ENTER**.

Note: To toggle between alarm history and active alarm display, press F3.

3. If this is a new robot with no gripper or End of Arm Tooling (EOAT), you must disable the hand broken check (error **SRVO-006 Hand Broken**). To do this,

- a. Press **MENU**, select 0 (Next), and select (SYSTEM), and then select **CONFIG**.

- b. Use the teach pendant arrow keys to move the cursor to the Hand Broken option, and press **ENTER**.

- c. Use the teach pendant keys to disable the hand broken setting for each motion group.

Note: It might already be disabled on your system.

MENU



MENU	2	SYSTEM	1
1 SELECT		1 Clock	
2 EDIT		2 Variables	
3 DATA		3 OF Release	
4 STATUS		4 Axis Limits	
5 4D GRAPHICS		5 Config	
6 SYSTEM		6 Motion	
7 USER2			
8 BROWSER			
9			
0 NEXT			

System/Config		42/59
41	Signal if OVERRIDE = 100	DO[0]
42	Hand broken :	<*GROUPS*>
43	Remote/Local setup:	OP panel key
44	External I/O(ON:Remote):DI	[0]
45	UOP auto assignment:	Full
46	Multi Program Selection:	TRUE
47	WAIT at Taught Position:	TRUE
48	Brake control ECO mode:	FALSE
49	J7,J8 jog key Setup :	<*DETAIL*>
50	Collection name(F1):	[STYLE]
51	Collection name(F2):	[COL]

4. If the error **MCTL-013 ENBL input is off** is displayed in the alarm history when you try to move the robot (it will not display as an active alarm) you must disable UI. To do this,

- a. Press **MENU**, select 0 (Next), and select **SYSTEM**, and then select **CONFIG**.

- b. Use the teach pendant arrow keys to move the cursor to the **Enable UI signals** option.

- c. Use the teach pendant keys to set the option to **FALSE**.

System/Config		7/59
1	Use HOT START:	FALSE
2	I/O power fail recovery:	RECOVER ALL
3	COLD START Autoexec program:	[*****]
4	HOT START Autoexec program:	[*****]
5	HOT START done signal:	DO[0]
6	Restore selected program:	TRUE
7	Enable UI signals:	FALSE
8	START for CONTINUE only:	FALSE
9	CSTOPI for ABORT:	FALSE

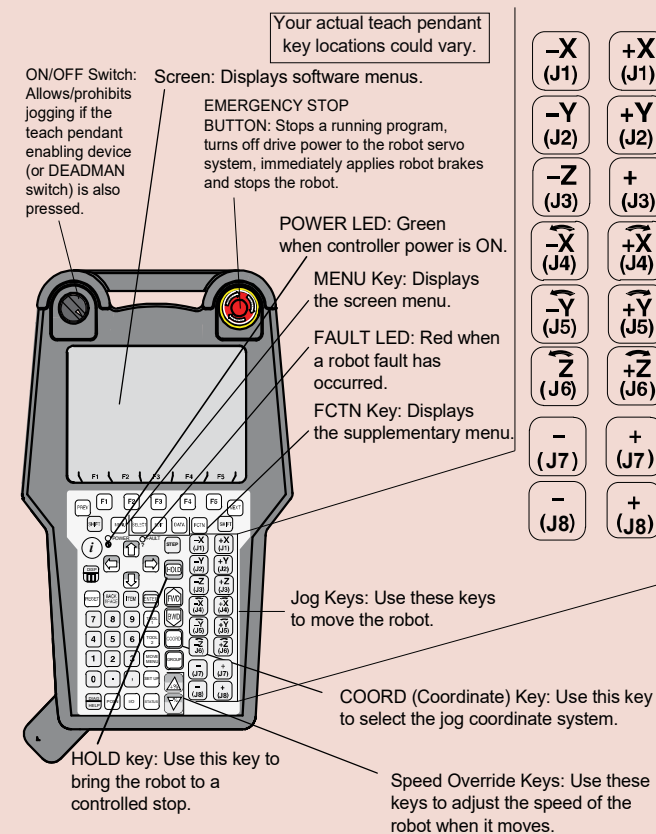
5. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

6. While pressing and holding a DEADMAN switch on the back of the teach pendant in its middle position, press **RESET** on the teach pendant to clear alarms.

7. To move the robot in cartesian coordinates (XYZ) press and hold **SHIFT** and press the X, Y, or Z jog key one at a time and watch the robot move. Refer to the **Robot (mechanical unit)** diagram on Page 1 of this guide.

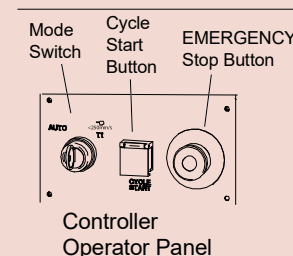
8. Ensure you are able to jog each robot joint.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the *Controller-specific Software Installation* manual.

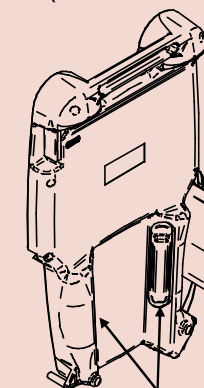


Refer to your *Application-Specific Setup and Operations Manual* for more information on each teach pendant key.

iPendant (Teach Pendant)



Controller Operator Panel



Teach Pendant Enabling Device (also called DEADMAN switch).

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to <https://www.fanucamerica.com/qr/edocs>.



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